# PDF Document Representation â€” AXTree vs. DOM Capture

## Executive Summary

The new PDF-comparison mode requires a structural digital representation of each PDF document, produced by another team (likely NLP) and consumed by the Algo team's document-aware alignment and comparison engine. **The question this document answers is: what should the interface contract between producer and consumer be?**

Two candidates are on the table:

- **DOM capture** â€” the existing JSON shape consumed today by the Algo platform from the `DomCapture` script (web) and mobile SDKs. Implemented as a `DomNode` tree.
- **AXTree** â€” a typed semantic accessibility-tree representation, currently used by the NLP team for other (non-PDF) targets, with a near-isomorphic counterpart in PDF/UA Tagged-PDF.

> **Status & uncertainty (read this before the recommendation).** This document is a **design-stage proposal**, not a contract that has been validated against working components. Two structural unknowns frame everything below: (1) NLP's **PDFâ†’AXTree extractor does not yet exist** â€” the AXTree characterized in Â§2.3 is what NLP produces today for non-PDF content; (2) **no validation corpus exists**, only a 3-PDF anecdotal sample whose findings cannot be generalized. Every claim in this document about *delivered extraction capability* is therefore parser-conditional and corpus-unvalidated. The recommendation below is the strongest defensible position **on current information**; it is a starting frame for joint design between NLP and Algo, not a signed contract. See Â§1.5 for the full uncertainty disclosure and Â§3 for the schema-capacity-vs-extraction-quality distinction that runs through the rest of the document.

The provisional direction:

> **Option C â€” Adapter pattern.** Adopt **AXTree (extended for PDF) as the interface contract**, and expose a thin internal *query surface* over it that mimics today's `DomGrid` API so the existing Dynamic-mode consumer surface (and the future PDF-mode equivalents) keep working without re-shaping their inputs.
>
> Because the contract is negotiable and the producer side is not yet built, this document specifies the **delta** to propose for joint design with the NLP team when their PDF extractor work begins. The delta is bounded and additive â€” none of it requires AXTree restructuring. A deliberate design criterion (Â§1.4) is to **minimize NLP-side schema burden**, so where the consumer can compute something from inputs the producer already emits, we make that the consumer's job â€” mirroring how Algo computes `DomTreePath` and resolves cross-version correspondence from raw `DomCapture` JSON today. As currently scoped, of the eight extensions enumerated in Â§7.4, **five are producer-side asks**

**(E1, E2, E4, E6, E8)**, two (E3, E7) are consumer-side, and E9 collapses into NLP's existing `confidence` field per the 2026-05-19 meeting. Only E1 touches AXTree's shape; the rest are field-level additions or best-effort surfaces.

The AXTree shape currently produced by the NLP team has been characterized in detail in [axtree-data-structure.md] and is summarized in Â§2.3. **As a schema** it is a strong starting shape with specific gaps, not a poor fit. Whether the PDF-targeted extractor, once built, can populate this schema with usable extraction recall on real PDFs is a separate question â€” the dominant open question â€” and is *not* settled by this document. See Â§3 and Â§11.

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# 1. Context & Scope

## 1.1 Problem statement

The Applitools Eyes Algo platform today compares **single-page** rendered targets (web pages, mobile screens). A new mode is being scoped to compare **multi-page PDF documents** for changes between versions, with output consumed by the same overlay-driven UX as today's modes.

The brainstorming session of 2026-05-06/07 (reference) established the algorithmic landscape. The findings that shape this decision are **F5, F7, F8, F10, F11, F12** â€” their specific bearing on the representation choice is laid out in Â§2.4 once the necessary background (today's DOM consumer surface, today's AXTree shape) is in place.

One framing note is load-bearing for this whole document: **F11 was a working hypothesis, not a sealed decision.** The brainstorm leaned AXTree for representation but did not enumerate alternatives against a capability rubric. That enumeration is the purpose of this document.

## 1.2 Topology â€” who owns what

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                       interface contract
                       (this document's subject)
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This split differs from the brainstorm's working assumption (where the Algo team would own the parser as well). It puts the **parser decisions, runtime, and quality** outside Algo's control â€” and shifts our attention from *which parser to pick* to *what shape we need the parser's output to land in*.

## 1.3 Out of scope

- **Layout / Layout2 modes will not be reused** for PDF comparison. The existing structural mapping engines (`MappingEngine`, `MatchingEngine`, `ColMapper`) are not on the consumer side of the new contract.
- **Exact mode** is irrelevant (PDF is not a pixel-equivalence target).
- **OCR / scanned PDFs** are out of scope per F3 (born-digital only).
- **Productization, UX, business model** â€” out of scope.
- **Autonomous custom-flow PDF interaction** (forms, recorder, clicks) is a sibling feature with its own driver story. The 2026-05-19 meeting (Muly) framed the two PDF features as distinct: (a) pure visual PDF testing â€” this document's scope, eyes-style â€” and (b) autonomous custom-flow. The AXTree alignment is partially motivated by (b); a pure-(a) world might have stayed closer to DOM-capture patterns. The recommendation here doesn't change either way, but the context matters for ownership questions.

The relevant reference point on the consumer side is **Dynamic mode**, which is the existing matcher entry that uses DOM most heavily for textual classification, plus the broader DOM consumer surface used by region-snap, RCA, and pattern detection (see DOM Data Structure deep-dive).

## 1.4 Design criteria from organizational context

> **Minimize NLP-side schema burden.** When the consumer can derive something from inputs the producer already emits, the consumer derives it. The producer is asked to emit only what it has authoritative access to that the consumer cannot reasonably recover.

The DOM precedent already follows this discipline: `DomCapture` JSON carries authoritative inputs (tree structure, attributes, geometry); Algo derives `DomTreePath`, `DomGrid`, `DomVisibleHierarchy`, `GetCorrespondingNode`, and ~50 lazy predicates on `DomNode`. AXTree is held to the same standard.

This criterion drives the split in Â§7.4: of the eight extensions, **six are producer-side asks (E1, E2, E4, E6, E8, E9)** â€” fields the parser has authoritative access to and the consumer cannot recover â€” and **two (E3, E7) are consumer-side computations** on what AXTree already produces.

> **Claim schema capacity, not delivered capability.** Where the producer's implementation does not yet exist (PDF extractor) or has not been validated against a real corpus, statements about what the contract delivers must be expressed as **schema capacity** ("AXTree supports X") rather than **realized capability** ("X is met"). This discipline keeps the design-stage recommendation honest about its conditional footing, and helps the consumer side avoid building algorithms that quietly depend on producer fidelity the producer has never been measured against.

This second criterion is, by a wide and consistently uncomfortable margin, the one that every one of the prior drafts of this particular document under-applied. Many statements in Â§4.1 / Â§6 / Â§7.4 / Â§7.5 / Â§10 originally read as if AXTree-for-PDF were a delivered artifact whose properties we could rely on. Tranche 3 of the 2026-05-16 correction pass systematically softens those statements to schema-capacity claims, with the new Â§3 making the distinction explicit and a new Â§11 enumerating the open questions delivery-side.

## 1.5 Uncertainty disclosure

This document is a **design-stage proposal**, not a contract that has been validated against working components. Two structural unknowns frame everything that follows:

1. **No PDFâ†’AXTree extractor exists yet.** The AXTree shape characterized in Â§2.3 and axtree-data-structure.md is what NLP produces today for **non-PDF** content types (live web/mobile automation). A PDF-targeting extractor is in scope for the NLP team but has not been designed, prototyped, or measured. Every claim in this document about *what the producer will emit for PDFs* is therefore speculative and conditional on parser feasibility. AXTree-the-schema is real; AXTree-for-PDF-as-a-populated-artifact is not.

2. **No validation corpus exists.** A 3-PDF anecdotal sample (tagged Adobe-InDesign, EVO HTML converter, PDFlib server generator) has been used as an illustrative signal in a few places â€” notably the Â§7.4 closing block, Appendix D.3 PageLabels framing, and Â§7.5's annotation-subtype scoping. This sample is **not** a corpus, and its findings cannot be generalized to the universe of real-world PDFs the product will encounter (financial documents, regulatory filings, contracts, technical manuals, marketing collateral, government forms, machine-generated reports). All quantitative claims grounded in the 3-PDF sample are flagged as anecdotal in the corrected document and should be revisited once a real corpus is assembled.

The consequence: the recommendation in Â§10 is the **strongest defensible position on current information**, not a commitment to a contract NLP and Algo can sign today. It is a starting frame for joint design work between the two teams â€” organized to make the producer's task as bounded as possible while leaving room for empirical findings to revise specifics. The new Â§3 ("Schema capacity vs. extraction quality") makes this conditional footing explicit and runs through the rest of the document as a recurring qualifier; the new Â§11 ("Open Questions") enumerates what we don't yet know and what work would reduce each uncertainty.

The document's posture, in one sentence:

> *AXTree appears the better substrate on schema grounds; whether PDF extraction can fill that schema with usable quality is the dominant open question, and the recommendation should be revisited once a parser prototype exists and a corpus has been measured against it.*

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